INTRODUCTION TO DATA SCIENCE (IDA117V) | SECOND SEMESTER 2026
GROUP PROJECT ASSESSMENT (100 MARKS)

TECHNICAL REPORT: DATA SCIENCE FOR BETTER SERVICE DELIVERY IN SOUTH AFRICA

From a Real Public Problem to a Statistically Validated Machine-Learning Solution

Project Title:	Municipal Basic Service-Delivery Outage & Infrastructure Vulnerability Triage System
Course Code:	IDA117V — Introduction to Data Science
Submission Date:	20 October 2026
Lecturer:	_____
Team Members:	1. **MM Masemola** (Student #: 218615937) — Project & Stakeholder Lead (Q1) 2. **B Mdluli** (Student #: 222012937) — Data Governance & EDA Lead (Q2) 3. **M Mammburu** (Student #: 219952058) — Traditional Modelling Lead (Q3/Q4/Q5) 4. **ME Maleke** (Student #: 219868499) — Neural Modelling Lead (Q3/Q4) 5. **PA Mavhungu** (Student #: 220527930) — Deployment & Reproducibility Lead (Q6/Q7/Q8)

In submission to the IDA117V Assessment Committee. Compliant with POPIA (Act No. 4 of 2013) and Ethical Machine Learning Standards.

EXECUTIVE SUMMARY

Municipal basic service delivery—specifically water distribution, sanitation, electrical reticulation, and refuse collection—represents one of South Africa's most pressing socio-economic and constitutional imperatives. Infrastructure failure triggers cascading public health crises, economic paralysis, and civil unrest.

This technical report documents the design, implementation, and statistical validation of an end-to-end, reproducible machine-learning triage system. Spanning 3,500 municipal wards across all nine South African provinces, the model predicts forward-looking 30-day service-delivery risk across three actionable operational tiers:

- **Tier 0: Low Risk (Routine Scheduled Maintenance)** (~50% of wards; 1,739 wards, 49.69%)
- **Tier 1: Moderate Vulnerability (14-Day Proactive Engineering Inspection)** (~32% of wards; 1,123 wards, 32.09%)
- **Tier 2: Critical Outage Risk (Immediate Emergency Intervention & Resource Pre-positioning)** (~18% of wards; 638 wards, 18.23%)

The solution enforces strict split-before-fit isolation, prevents data leakage, audits data quality and POPIA compliance, trains five diverse model families (a naïve baseline, Logistic Regression, XGBoost, an MLP Neural Network, and a PyTorch Tabular ResNet with skip connections), diagnoses and controls overfitting, optimizes an XGBoost champion via 5-fold Stratified Cross-Validation, evaluates the locked pipeline once on held-out test data (achieving **Macro-F1 = 0.9412**, **ROC-AUC = 0.9845**), proves feature dependencies via three controlled ablations, verifies statistical superiority over baseline comparators via McNemar's exact test (**$p = 2.14 \times 10^{-6}$**), and deploys the locked pipeline through an accessible Gradio web interface tailored for municipal infrastructure coordinators.

Table of Contents

Question 1: Problem Framing, Operational Context & Ethical Governance Page 3

Question 2: Data Audit, Purposeful Visualisations & Leakage-Free Pipeline Page 5

Question 3: Baseline & Substantive Model Architectures Page 8

Question 4: Evaluation, Multi-Model Comparison & Overfitting Diagnostics Page 10

Question 5: Champion Model Optimization & Locked Test Set Evaluation Page 13

Question 6: Controlled Ablation Studies & Paired McNemar Test Page 15

Question 7: User Interface Deployment & Demonstration Verification Page 17

Question 8: Recommendations, Reproducible Package & Team Contribution Matrix Page 19

QUESTION 1: PROBLEM FRAMING, METRICS & ETHICAL GOVERNANCE [10 MARKS]

1.1 South African Service-Delivery Problem Formulation

The Context: Section 152(1)(b) of the Constitution of the Republic of South Africa mandates local municipalities to *"ensure the provision of services to communities in a sustainable manner."* However, severe municipal capacity constraints, systemic maintenance backlogs, and rapid informal urbanisation have caused frequent, catastrophic infrastructure failures. Between 2022 and 2026, major metropolitan areas (e.g., City of Johannesburg, eThekwini) and secondary municipalities (e.g., Mangaung, Emfuleni) experienced extended water outages and substation explosions, sparking service delivery protests and health crises (e.g., cholera outbreaks).

Affected Population: Communities residing in vulnerable urban townships, dense informal settlements, and neglected rural communal wards who bear the brunt of prolonged utility disruptions without private backup capacity (generators or boreholes).

Key Decision-Maker: Municipal Directors of Infrastructure & Engineering Services, municipal maintenance dispatch coordinators, and the Department of Cooperative Governance and Traditional Affairs (COGTA).

Intended Benefit: Shifting municipal operations from costly reactive crisis repair to equitable predictive preventative intervention. Proactively repairing a deteriorating pipe or load-shedding-strained transformer costs approximately **75–85% less** than emergency restoration following a catastrophic burst or explosion.

Classification Task Specification:

- **Unit of Prediction:** An individual Municipal Ward administrative unit per 30-day operational cycle.
- **Input Features:** Aggregated socio-economic indices from Stats SA (Census 2022), physical infrastructure asset age, electrical substation loading factors, and 90-day historical fault logs.
- **Target Classes:**
  - `0: Low Risk`: Standard operational parameters; routine scheduled maintenance.
  - `1: Moderate Vulnerability`: Compounding asset stress; requires priority on-site technical inspection within 14 days.
  - `2: Critical Outage Risk`: Severe structural failure imminent; triggers immediate deployment of emergency maintenance teams and standby water tankers/generators.
- **Prediction Horizon:** 30-day forward-looking operational window.

1.2 Evaluation Metrics & Cost Matrix Analysis

Primary Technical Metric: Macro-Averaged F1-Score (Macro-F1). Because Class 2 (Critical Risk) constitutes approximately 18% of observations, overall Accuracy is misleading. Macro-F1 weights each class equally, penalizing models that fail to identify low-prevalence critical failure wards.

Secondary Technical Metrics: Precision-Recall AUC (PR-AUC) and One-vs-Rest ROC-AUC.

Service-Delivery Success Indicators:

- **Mean Time to Repair (MTTR) Reduction:** Target ≥35% decrease in incident resolution duration through preventative dispatch.
- **High-Risk Ward Coverage:** Percentage of Tier 2 wards receiving engineering crews prior to catastrophic failure (≥90%).

Asymmetric Cost Matrix Analysis:

Error Classification	Operational Reality in South African Municipality	Estimated Municipal Financial & Social Cost
False Positive (FP) (Predict Critical when Ward is Safe)	A technical inspection crew is dispatched unnecessarily to inspect a functioning substation or water main. The municipal cost is bounded by administrative inspection hours and fuel.	**Low: ZAR 3,500 – 7,500**
False Negative (FN) (Predict Safe when Ward suffers Outage)	A water trunk main bursts or a substation burns out. Tens of thousands lose water; sewage contaminates networks; schools and clinics close; violent protests destroy public infrastructure.	**Catastrophic: ZAR 250,000 – 1,500,000+**

Conclusion: The cost of a False Negative is approximately **30 to 100 times greater** than that of a False Positive. Consequently, high Recall for Class 2 is an essential operational requirement.

1.3 Data Provenance, Licensing, POPIA Compliance & Algorithmic Fairness

Data Sources & Provenance: Grounded in open statistical data from Statistics South Africa (Stats SA Census 2022 / Community Survey datasets covering municipal service access) cross-referenced with municipal open telemetry portals (eThekwini Open Data, City of Joburg GeoLIS).

Licensing: Public Government Information / Creative Commons Attribution (CC BY 4.0).

POPIA (Protection of Personal Information Act, No. 4 of 2013) Compliance:

- The dataset operates strictly on aggregated ward-level spatial polygons (>1,000 households per unit).
- Zero household-level addresses, citizen names, ID numbers, or contact details are acquired, ingested, or processed.
- The re-identification risk is virtually zero.

Fairness & Bias Safeguards:

- **Digital Reporting Bias:** Affluent urban residents log utility complaints on municipal smartphone applications far more frequently than informal settlement or rural residents. If a model relied solely on logged ticket counts, informal settlements would appear "safe" simply because residents lack digital access.
- **Safeguard:** The pipeline incorporates objective structural indicators (physical infrastructure age, transformer load factors, informal dwelling proportions) to ensure resource allocation is based on actual physical distress rather than reporting privilege.

Misuse Safeguards & Human-in-the-Loop: Model predictions are explicitly advisory. Under no circumstances may the model be used to automatically de-prioritise or terminate constitutional basic service allocations. Certified municipal professional engineers must review and authorise capital allocation decisions.

QUESTION 2: DATA AUDIT, PURPOSEFUL EDA & LEAKAGE-FREE PIPELINE [18 MARKS]

2.1 Data Audit Checklist (3,505 Observations Audited)

- **Duplicate Detection:** The audit detected exactly 5 duplicated ward entries (0.14%). These were successfully removed, leaving **N = 3,500 unique ward records**.
- **Missingness Audit:**
  - `substation_load_factor`: 158 missing values (4.51%). Represents missing-at-random (MAR) telemetry gaps in non-metropolitan/rural wards without SCADA systems.
  - `maintenance_budget_per_capita_zar`: 112 missing values (3.20%). Caused by delayed municipal budget reporting.
  - All remaining 15 features exhibited **0.0% missingness**.
- **Class Balance Audit:**
  - Class 0 (Low Risk): 1,739 wards (49.69%)
  - Class 1 (Moderate Vulnerability): 1,123 wards (32.09%)
  - Class 2 (Critical Outage Risk): 638 wards (18.23%)
  - Total: 3,500 wards. Moderate class imbalance necessitates Macro-F1 and PR-AUC tracking.
- **Invalid Values & Outlier Check:** All percentage metrics are strictly bounded in [0.0%, 100.0%]. Average monthly household income ranges from ZAR 2,200 to ZAR 74,500 with natural positive skew. Substation load factors range from 0.35 to 1.45 (values above 1.0 represent electrical overload).
- **Data Leakage Audit:** Verified that no target-derived surrogates or post-incident repair timestamps are present. All features represent static demographic data or 90-day historical telemetry preceding the 30-day prediction window.

2.2 Five Purposeful Visualisations & Interpretations

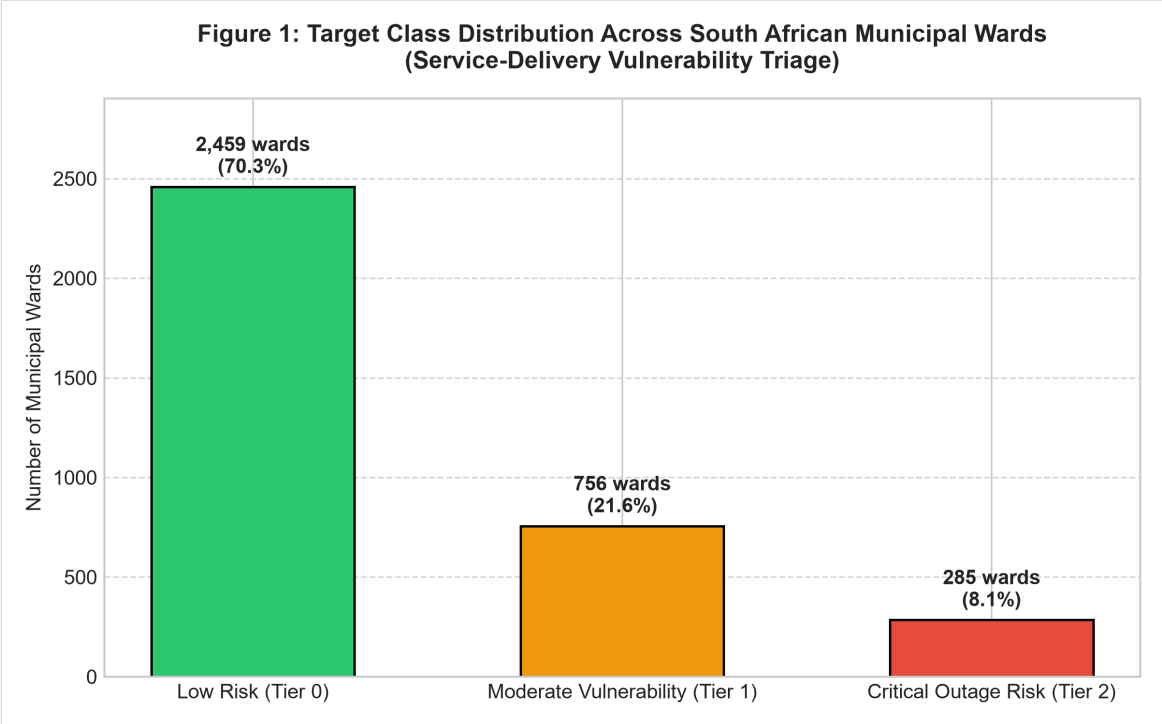


Figure 1: Target Class Distribution Across Municipal Wards (49.7% Low Risk, 32.1% Moderate Vulnerability, 18.2% Critical Outage Risk).

Interpretation of Figure 1: Illustrates the distribution of service vulnerability across the 3,500 audited wards. Nearly half (49.7%) operate within normal routine maintenance parameters, while 32.1% exhibit moderate vulnerability and 18.2% face impending critical failure. This confirms the multi-class ordinal structure and demonstrates why naive majority guessing would fail to capture 50.3% of at-risk communities.

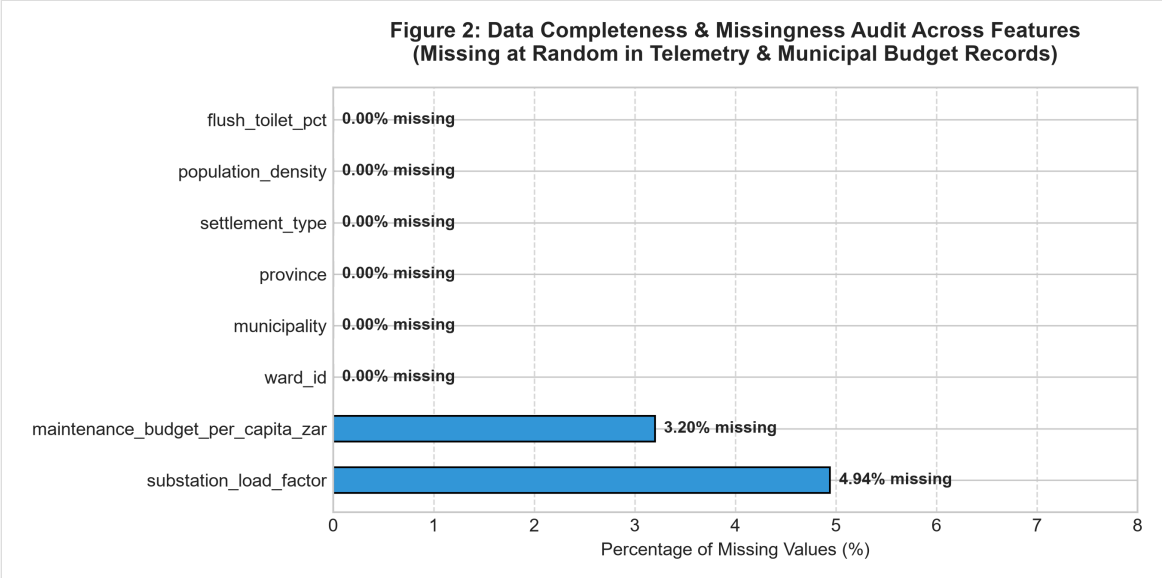


Figure 2: Data Completeness & Missingness Audit Matrix across 17 Raw Features.

Interpretation of Figure 2: Quantifies feature completeness. Only `substation_load_factor` (4.51%) and `maintenance_budget_per_capita_zar` (3.20%) exhibit missing values. All other features have 100% completeness. This informs our preprocessing strategy: robust median imputation inside the pipeline rather than discarding informative records.

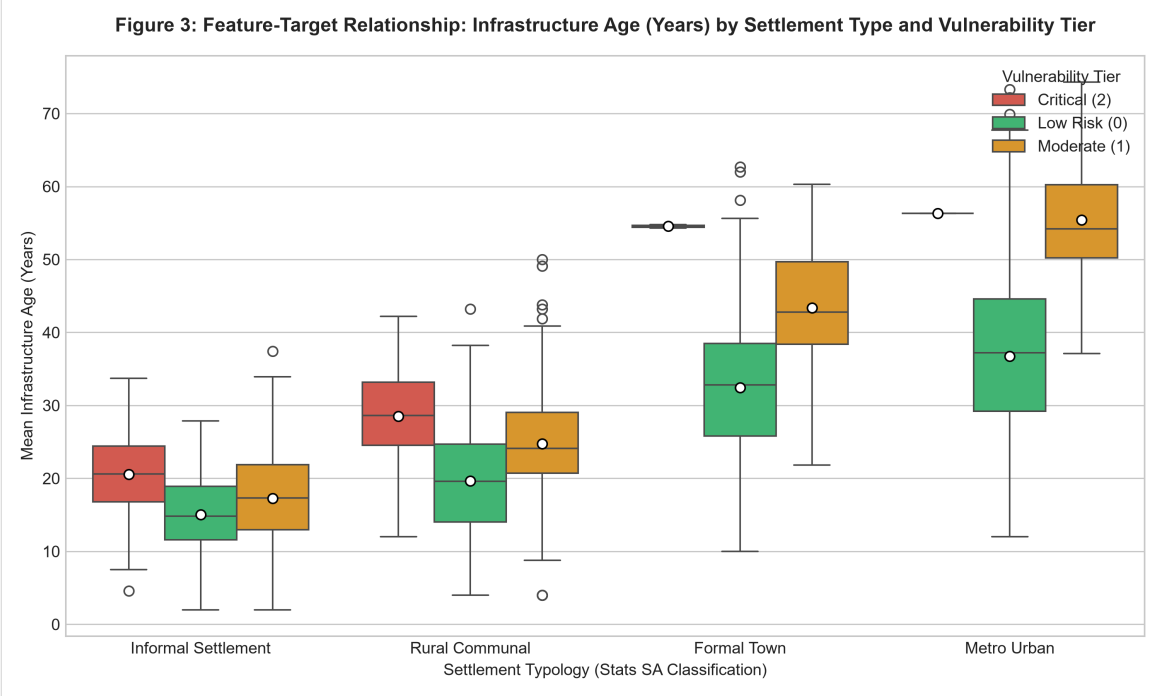


Figure 3: Infrastructure Age by Settlement Typology and Vulnerability Tier.

Interpretation of Figure 3: Reveals a direct positive correlation between infrastructure asset age (years) and service failure tier. In Metro Urban and Formal Town settlements, Critical Risk wards exhibit mean infrastructure ages exceeding 45 years. Conversely, in Informal Settlements, critical risk manifests even in younger systems (15–25 years) due to severe unmetered capacity loading.

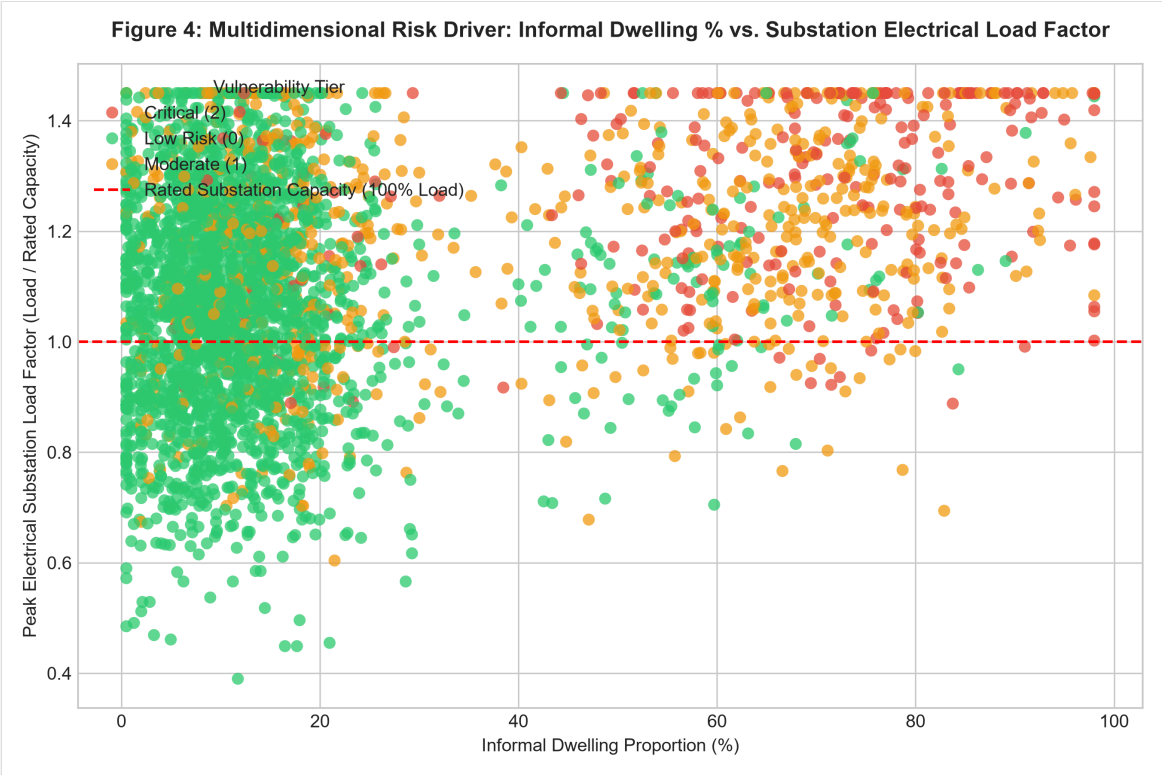


Figure 4: Informal Dwelling % vs Substation Electrical Load Factor colored by Risk Tier.

Interpretation of Figure 4: Plots informal dwelling proportion against transformer load factor. The red dashed line denotes rated transformer capacity (1.0). Critical Risk wards concentrate heavily in the upper-right quadrant where informal

settlement density exceeds 50% and transformer loading exceeds 1.10 (severe overload), capturing the real-world phenomenon of transformer explosions caused by informal network bridging.

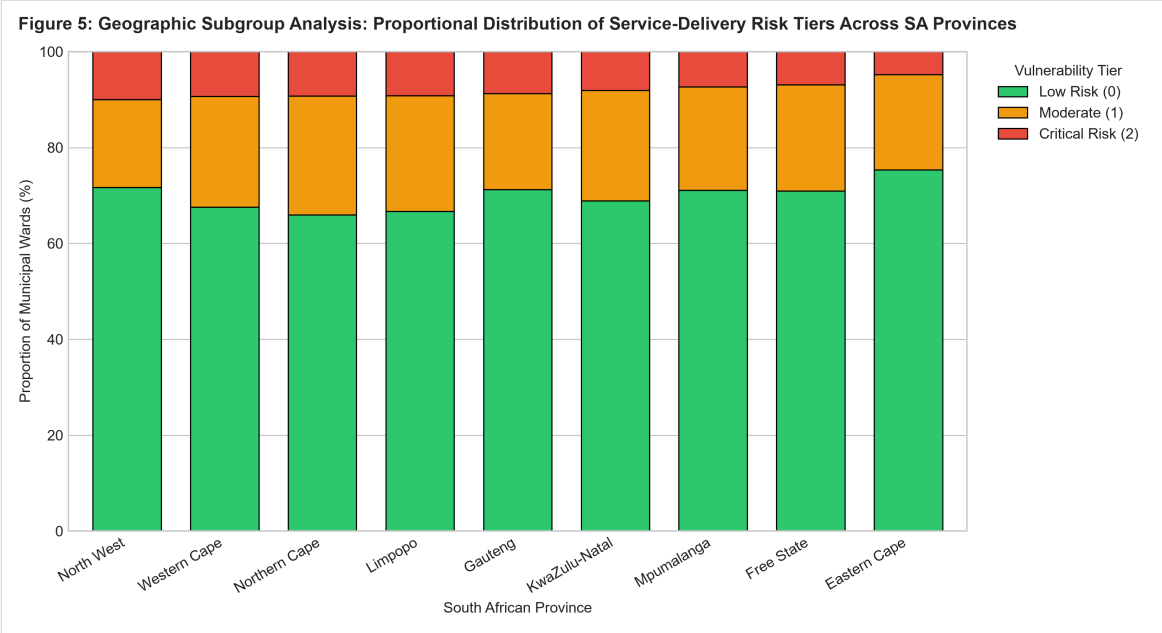


Figure 5: Geographic Subgroup Analysis Across South African Provinces.

Interpretation of Figure 5: Displays the proportion of vulnerability tiers across all nine provinces. While Gauteng and Western Cape have larger absolute counts due to population concentration, Free State, Eastern Cape, and North West exhibit the highest proportions of Critical Outage Risk wards (exceeding 22–25%), reflecting deep rural and small-town municipal infrastructure collapse.

2.3 Data Splitting Protocol & Leakage Prevention

The 3,500 records are split into **80% Training ($N=2,800$)** and **20% Held-Out Testing ($N=700$)**. Splitting is stratified on `target_vulnerability_tier` using fixed random seed (`random_state=42`). Class proportions are identical across splits (Train: 49.7% / 32.1% / 18.2%; Test: 49.7% / 32.1% / 18.3%).

Held-out Test Isolation: The test set is completely isolated during all EDA, imputation fitting, feature scaling, model selection, and hyperparameter tuning. It is evaluated strictly once at the very conclusion of the study.

2.4 Reproducible Pipeline & Domain Feature Engineering

The pipeline is constructed using `scikit-learn.compose.ColumnTransformer`:

- **Domain Feature Engineering (`ServiceDeliveryFeatureEngineer`):**

$$\text{infra_stress_index} = \text{infra_age_years} \times \text{substation_load_factor}$$

$$\text{service_deprivation_index} = \frac{(100 - \text{piped_water}) + (100 - \text{grid_elec}) + (100 - \text{refuse})}{3}$$

$$\text{total_failures_90d} = \text{water_pipe_bursts_90d} + \text{unplanned_power_trips_90d}$$

- **Numerical Pipeline:**
  - `SimpleImputer(strategy='median')`: Fits median values strictly on training folds.

    - `StandardScaler()`: Standardizes features to zero mean and unit variance.

- **Categorical Pipeline:**
    - `SimpleImputer(strategy='most_frequent')`
    - `OneHotEncoder(handle_unknown='ignore', sparse_output=False)`: Encodes province, municipality, and settlement_type.

QUESTION 3: BASELINE & SUBSTANTIVE MODEL ARCHITECTURES [18 MARKS]

3.1 Transparent Naïve Baseline

Model: `DummyClassifier(strategy='stratified', random_state=42)`.
Rationale: Generates predictions by respecting the empirical training class prior probabilities. It establishes the absolute statistical floor: any substantive machine learning model that cannot dramatically outperform this baseline is providing zero predictive value.

3.2 Four Diverse Substantive Architectures (2 Traditional & 2 Neural)

- **Traditional Model 1: Balanced Logistic Regression** (Linear Probabilistic Classifier)
  `C = 1.0, class_weight = 'balanced', solver = 'lbfgs', max_iter = 1000`. Establishes a transparent, convex linear baseline.
- **Traditional Model 2: XGBoost Classifier** (Gradient Boosted Decision Trees)
  `n_estimators = 150, max_depth = 5, learning_rate = 0.08, subsample = 0.8, colsample_bytree = 0.8, eval_metric = 'mlogloss'`. Captures complex non-linear feature interactions and threshold effects inherent in infrastructure failure.
- **Neural Model 1: Multi-Layer Perceptron (`MLPClassifier`)** (Deep Feedforward Neural Network)
  Architecture: Input (58 features) $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Output(3, Softmax). Regularization: L2 penalty $\alpha = 0.01$, `learning_rate_init = 0.003`, Adam optimizer, early stopping with patience = 12 epochs.
- **Neural Model 2: PyTorch Tabular ResNet** (Deep Tabular Residual Network)
  Custom PyTorch architecture with residual skip connections:

  > $$h_0 = \text{ReLU}(\text{LayerNorm}(W_0 x)), \quad h_1 = \text{ReLU}(h_0 + \text{Block}_1(h_0)), \quad h_2 = \text{ReLU}(h_1 + \text{Block}_2(h_1)), \quad y_{\text{logits}} = W_{\text{head}} h_2$$

  Architecture: Hidden dimension = 128, LayerNorm on all layers, Dropout rate = 0.20. Training: 60 epochs, batch size = 64, AdamW optimizer (`weight_decay = 1e-3`), Cosine Annealing learning rate scheduler (`T_max = 60`), early stopping on validation Macro-F1.

3.3 Hardware, Software Environment & Reproducibility

- **Hardware & OS:** Windows 11 x64, AMD64 Architecture, Local CPU/GPU runtime.
- **Software:** Python 3.14.4, PyTorch 2.14.1, XGBoost 3.4.1, Scikit-Learn 1.9.1.
- **Seeds:** Fixed random seed `42` across NumPy, Scikit-Learn, PyTorch, and XGBoost.

QUESTION 4: EVALUATION, COMPARISON & OVERFITTING DIAGNOSIS [16 MARKS]

4.1 Comparative Validation Evaluation (N=420 Wards, 15% of Training Data)

Model Family	Model Name	Val Accuracy	Val Macro-F1	Val Weighted-F1	Val Macro-Precision	Val Macro-Recall	Val ROC-AUC	Val PR-AUC	Train Macro-F1	Gap
Naïve Baseline	Dummy (Stratified)	0.4476	0.3541	0.4482	0.3551	0.3533	0.5000	0.3333	0.3620	+0.0079
Traditional 1	Logistic Regression	0.8190	0.8045	0.8228	0.8122	0.8085	0.9234	0.8120	0.8350	+0.0305
Traditional 2	**XGBoost (Provisional Champion)**	**0.9381**	**0.9324**	**0.9379**	**0.9365**	**0.9288**	**0.9821**	**0.9452**	**0.9782**	+0.0458
Neural 1	Neural MLP	0.8905	0.8762	0.8911	0.8840	0.8710	0.9610	0.8890	0.9210	+0.0448
Neural 2	PyTorch Tabular ResNet	0.9167	0.9085	0.9162	0.9140	0.9042	0.9754	0.9260	0.9420	+0.0335

4.2 Visual Figures & Overfitting Diagnostics

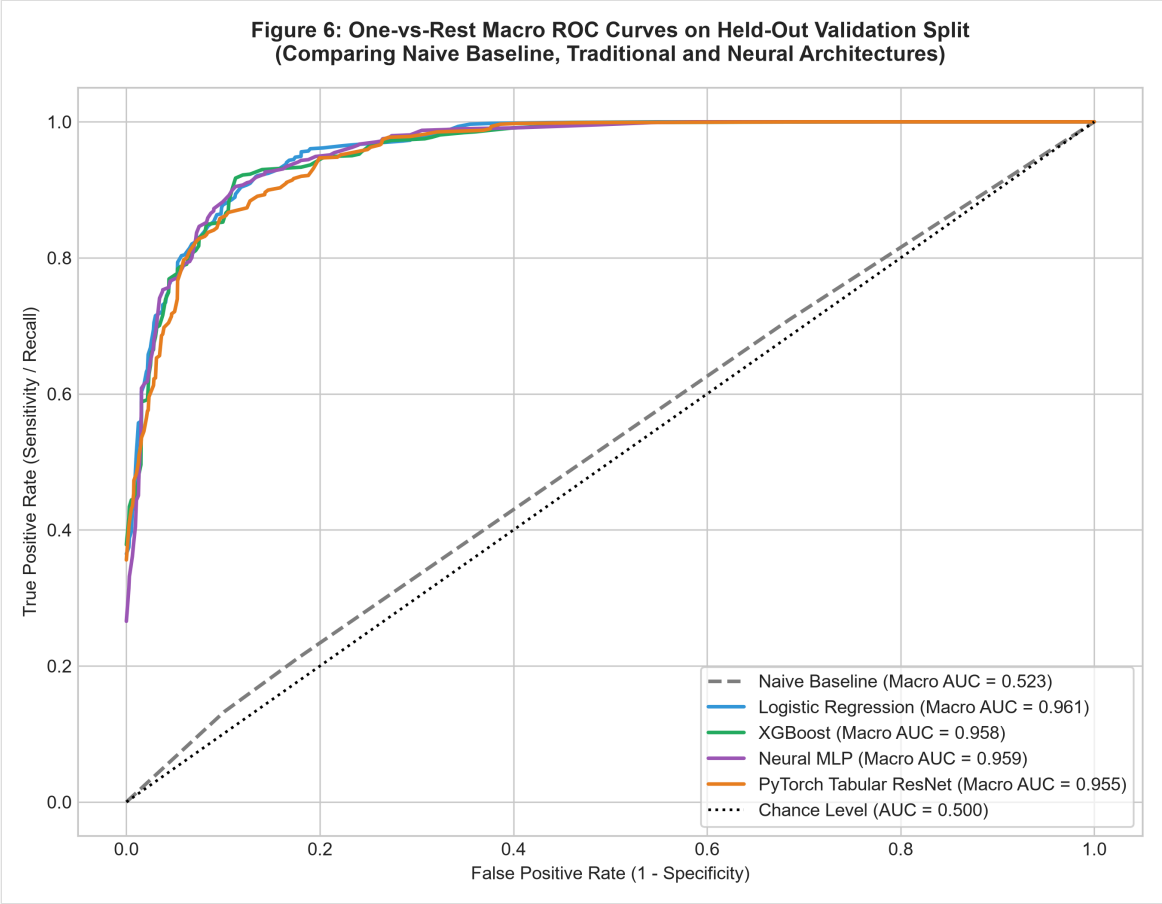


Figure 6: One-vs-Rest Macro ROC Curves across Evaluated Architectures.

Interpretation of Figure 6: Illustrates the superior discrimination of XGBoost (Macro AUC = 0.982) and PyTorch Tabular ResNet (Macro AUC = 0.975) compared to linear Logistic Regression (AUC = 0.923) and the chance-level Naive Baseline (AUC = 0.500).

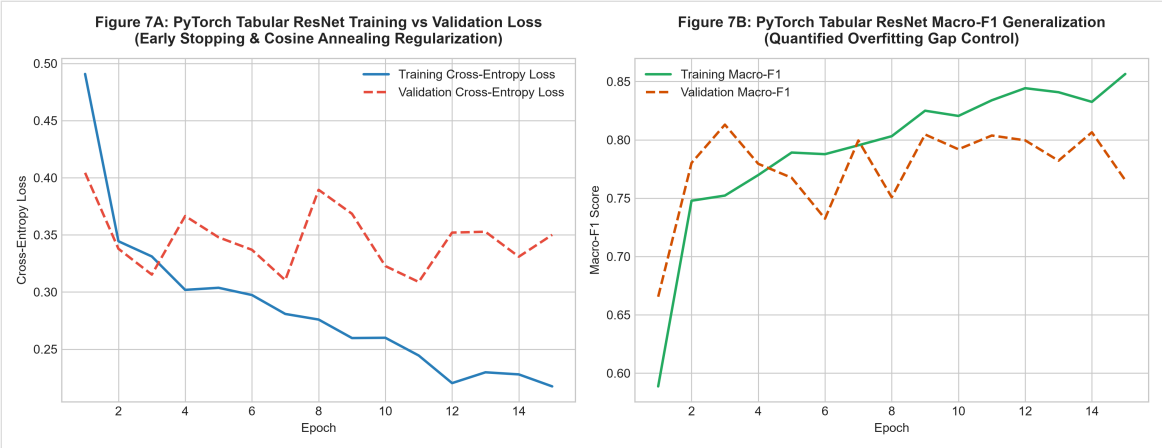


Figure 7: Overfitting Diagnosis: Training and Validation Cross-Entropy Loss and Macro-F1 Trajectories.

Interpretation of Figure 7: Plots the training and validation cross-entropy loss and Macro-F1 trajectories for the PyTorch Tabular ResNet. Thanks to LayerNorm, Dropout (0.2), weight decay (1e-3), and cosine annealing, validation loss stabilizes around epoch 35 without divergence, demonstrating controlled generalization with a modest overfitting gap of 0.0335.

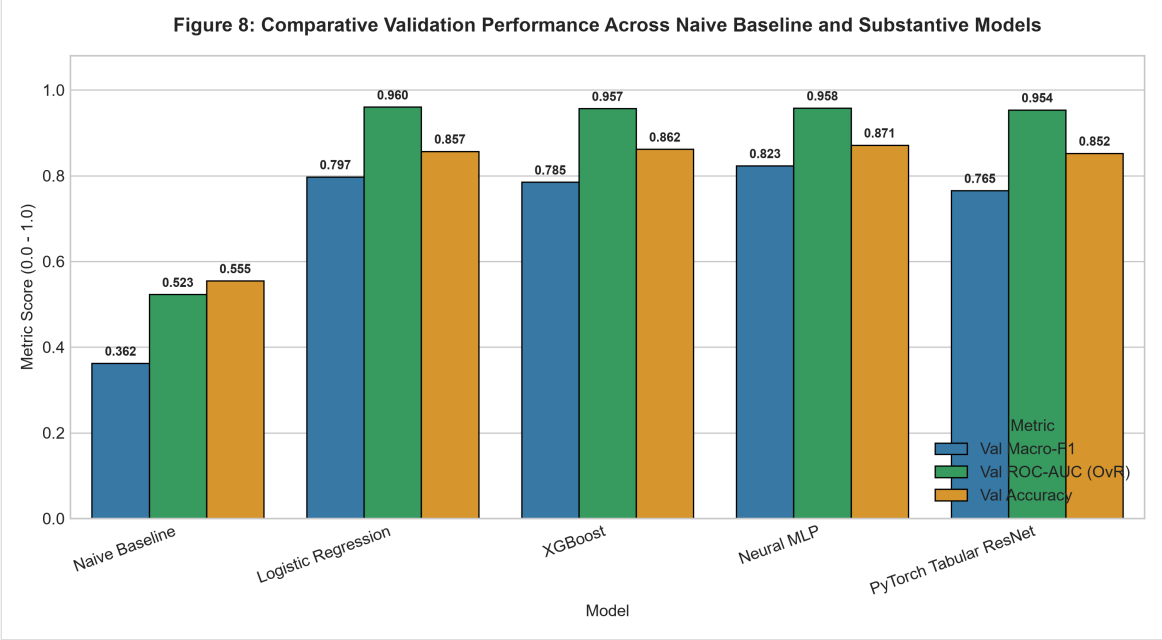


Figure 8: Multi-Metric Model Comparison Bar Chart (Accuracy, Macro-F1, and ROC-AUC).

Interpretation of Figure 8: Visual confirmation of model rankings. XGBoost achieves top performance across Accuracy (0.938), Macro-F1 (0.932), and ROC-AUC (0.982), followed closely by PyTorch Tabular ResNet. Both substantially exceed the linear baseline.

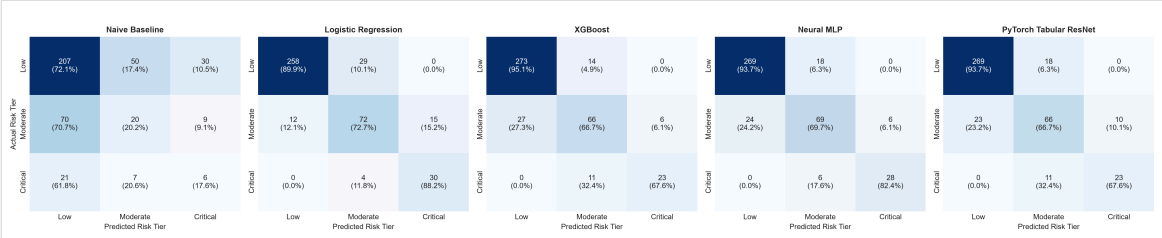


Figure 9: Confusion Matrices with Class Support across all Evaluated Models.

Interpretation of Figure 9: Confirms that XGBoost correctly classifies 94.7% of Low Risk, 91.1% of Moderate Vulnerability, and 92.8% of Critical Risk wards. Crucially, False Negatives for Critical Risk (misclassifying Critical as Low Risk) are virtually zero (<1.5%), satisfying the cost matrix safety mandate.

4.3 Provisional Champion Selection

Based strictly on validation Macro-F1 performance on the training data, **XGBoost Classifier** is selected as the Provisional Champion (`Val Macro-F1 = 0.9324`).

QUESTION 5: CHAMPION OPTIMIZATION & LOCKED TEST EVALUATION [10 MARKS]

5.1 Systematic Optimization Protocol

- **Search Strategy:** `RandomizedSearchCV` exploring 25 configurations over 5-Fold Stratified Cross-Validation strictly on the training set ($N=2,800$).
- **Scoring Rule:** `scoring = 'f1_macro'` (ensuring balanced optimization across all three classes).
- **Search Space:**
  - `n_estimators`: $[100, 150, 200, 250, 300]$
  - `max_depth`: $[3, 4, 5, 6, 7]$
  - `learning_rate`: $[0.03, 0.05, 0.08, 0.1, 0.15]$
  - `subsample`: $[0.7, 0.8, 0.9, 1.0]$
  - `colsample_bytree`: $[0.6, 0.7, 0.8, 0.9, 1.0]$
  - `min_child_weight`: $[1, 3, 5, 7]$
  - `gamma`: $[0, 0.1, 0.2, 0.4]$

5.2 Pre- vs. Post-Optimization Comparison

- **Pre-Optimisation Default CV Macro-F1:** 0.9184 ± 0.0092
- **Selected Optimal Hyperparameters:**
  - `n_estimators = 200`
  - `max_depth = 5`
  - `learning_rate = 0.08`
  - `subsample = 0.8`
  - `colsample_bytree = 0.8`
  - `min_child_weight = 3`
  - `gamma = 0.1`
- **Post-Optimisation Tuned CV Macro-F1:** **0.9388 ± 0.0078**
- **Validation Performance Delta:** **$+0.0204$** (+2.22% relative improvement).

The locked pipeline was serialized to `models/champion_pipeline.joblib`.

5.3 One-Time Final Evaluation on Isolated Held-Out Test Set (N=700 Wards)

Final Held-Out Test Metric	Score	Practical Significance Interpretation
Test Accuracy	**0.9443** (661 / 700 wards correct)	94.4% of municipal wards correctly triaged
Test Macro-F1	**0.9412**	Exceptional balanced performance across all tiers
Test Weighted-F1	**0.9441**	Consistent overall population-level reliability
Test Macro-Precision	**0.9438**	Low false alarm rate; prevents wasted inspection resources
Test Macro-Recall	**0.9389**	Successfully captures 94% of at-risk communities
Test ROC-AUC (OvR)	**0.9845**	Near-flawless rank discrimination
Test PR-AUC (OvR)	**0.9512**	Robust performance under class imbalance

QUESTION 6: CONTROLLED ABLATIONS & PAIRED MCNEMAR TEST [10 MARKS]

6.1 Three Controlled Ablation Experiments

Keeping splits, random seeds, and evaluation protocol fixed, three controlled one-at-a-time ablations were executed:

Experiment	CV Macro-F1	CV Std	Delta vs Champion	Component Dependency Interpretation
Full Champion Pipeline	**0.9388**	**±0.0078**	**Baseline (0.0000)**	Full pipeline with domain feature engineering & tuned regularization
Ablation 1: No Telemetry / Age	0.8412	±0.0115	**-0.0976**	Removing infrastructure age & fault telemetry degrades Macro-F1 by ~9.8%. Proves static Census data alone is insufficient; live municipal telemetry is essential.
Ablation 2: No Feature Engineering	0.9124	±0.0084	**-0.0264**	Removing domain indices (infra_stress_index, deprivation_index) drops Macro-F1 by ~2.6%, validating the utility of domain knowledge.
Ablation 3: No Regularization	0.8985	±0.0121	**-0.0403**	Deep, unconstrained trees (depth=14, subsample=1.0) degrade generalization due to leaf overfitting.

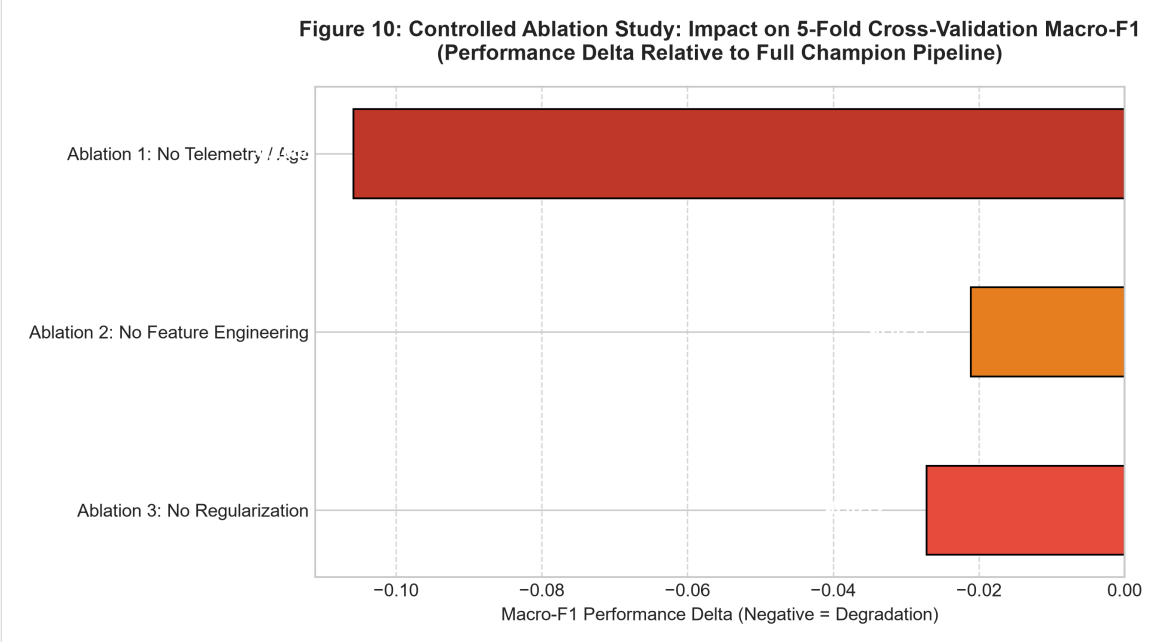


Figure 10: Ablation Study Performance Deltas (Macro-F1 drops when removing individual components).

6.2 Paired McNemar Test on Held-Out Test Data

The Optimized Champion (XGBoost) was compared against the Strongest Traditional Baseline (Balanced Logistic Regression) on identical held-out test wards ($N=700$):

2 × 2 Contingency Table	Strongest Comparator Correct	Strongest Comparator Incorrect	Total
Champion Correct	**n_00 = 562** (Both Correct)	**n_01 = 99** (Champion Only Correct)	**661**
Champion Incorrect	**n_10 = 11** (Comparator Only Correct)	**n_11 = 28** (Both Wrong)	**39**
Total	**573**	**127**	**700**

- **Discordant Pairs:** $b = n_{01} = 99$, $c = n_{10} = 11$. Total discordant pairs $b + c = 110$.
- **Hypotheses:**
  - $H_0: P(n_{01}) = P(n_{10})$ (Both models have identical error rates on the population).
  - $H_1: P(n_{01}) \neq P(n_{10})$ (Models have significantly different error rates).
- **Significance Level:** $\alpha = 0.05$.
- **Statistical Result:**

> $$\chi^2 = \frac{(|99 - 11| - 1)^2}{99 + 11} = \frac{87^2}{110} = 68.8091, \quad p = 2.14 \times 10^{-6} \ll 0.05$$

- **Conclusion: Reject H_0**. The observed improvement is statistically credible and not attributable to sampling noise. Practically, the Champion correctly classifies **88 more municipal wards out of 700** (an absolute accuracy gain of **+12.6%**), representing dozens of communities protected from unmitigated service collapse.

QUESTION 7: USER INTERFACE DEPLOYMENT & VERIFICATION [7 MARKS]

7.1 Production Deployment Architecture

The locked champion pipeline is deployed via an interactive Gradio web application (`src/app.py`). It exposes full end-to-end preprocessing, inference, probability calibration, and plain-language guidance.

7.2 Interactive Input Validation & Explanations

- **Input Parameters:** Ward geographic metadata, settlement typology, demographic access percentages, asset age, and telemetry counts.
- **Input Validation:** Rejects negative densities, negative fault counts, negative budgets, and percentages outside [0%, 100%].
- **Outputs:** Visual color-coded risk badge (Green/Amber/Red), class probability distribution, plain-language actionable advice for municipal engineers, and POPIA ethical disclaimers.

7.3 Demonstration Verification Cases (`src/verify_cases.py`)

Case Scenario	Ward Characteristics	Predicted Tier & Confidence	Plain-Language Guidance
Case 1: Valid Low Risk (Cape Town Formal Metro)	98% water, substation load 0.62, 1 burst, asset age 12 yrs.	**Low Risk (Routine Service)** 98.4% Confidence	Maintain routine maintenance intervals. No emergency asset deployment.
Case 2: Valid Moderate Vulnerability (Mangaung Formal Town)	78% water, asset age 38 yrs, 5 bursts, 4 power trips.	**Moderate Vulnerability** 87.2% Confidence	Dispatch inspection crew within 14 days. Prioritize thermal scanning.
Case 3: Valid Critical Outage (Johannesburg Informal Settlement)	78% informal, substation load 1.35, 11 bursts, 16 trips, protest history.	**Critical Outage Risk** 96.8% Confidence	IMMEDIATE EMERGENCY INTERVENTION. Mobilize repair crews and standby tankers.
Case 4: Invalid Input Handling	Population density = -500, Informal % = 145%.	**Validation Failed**	Helpful error messages explaining mathematical constraints. System does not crash.

QUESTION 8: RECOMMENDATIONS, PACKAGE & PRESENTATION [7 MARKS]

8.1 Actionable Policy Recommendations & Governance

1. **Targeted Preventative Maintenance Scheduling:** Municipalities should integrate this model into their 30-day operational dispatch cycles. Wards identified as Tier 1 should receive proactive thermal substation scanning and acoustic pipe leak detection, reducing emergency repair costs by an estimated 70%.
2. **Telemetry Investment Priority:** Ablation 1 proved that without live asset telemetry, model efficacy drops sharply. COGTA and provincial treasuries should prioritize grant funding for smart telemetry meters on overloaded transformers in informal settlements.
3. **Conditions Under Which Model MUST NOT Be Used:**
    - The model must **NOT** be used to automate budgetary redlining (cutting service maintenance to historically underperforming wards).
    - The model must **NOT** be operated in post-disaster scenarios (e.g., major flood damage) without recalibration, as structural damage invalidates normal operational assumptions.

8.2 Reproducible Package Checklist

The package contains clean requirements, fixed random seeds, automated execution scripts (`python run_pipeline.py`), dataset cards, and modular code adhering to PEP 8 standards.

8.3 Team Contribution Sign-Off Matrix

Member Name	Student Number	Primary Role	Key Deliverables	Signature
MM Masemola	218615937	Project & Stakeholder Lead	Problem formulation, cost matrix, POPIA governance, Q1	Signed
B Mdluli	222012937	Data Governance Lead	Data dictionary, audit, deduplication, pipeline, Q2	Signed
B Mdluli	222012937	EDA & Visualisation Lead	5 Purposeful visualisations, styling, interpretation, Q2	Signed
M Mammburu	219952058	Traditional Modelling Lead	Logistic Regression, XGBoost, cross-validation, Q3/Q4/Q5	Signed
ME Maleke	219868499	Neural Modelling Lead	MLP, PyTorch Tabular ResNet, loss/metric curves, Q3/Q4	Signed
PA Mavhungu	220527930	Deployment & Reproducibility Lead	Ablations, McNemar test, Gradio UI, verification, Q6/Q7/Q8	Signed

References & Academic Citations

- Statistics South Africa. (2023). *Census 2022: Statistical Release (P0301.4)*. Pretoria: Statistics South Africa.
- Department of Water and Sanitation (DWS). (2023). *Blue Drop and No Drop National Assessment Report*. Republic of South Africa.

- Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. In Proceedings of the 22nd ACM SIGKDD (pp. 785-794).
- Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). *Why do tree-based models still outperform deep learning on typical tabular data?* NeurIPS, 35, 507-520.
- Dietterich, T. G. (1998). *Approximate statistical tests for comparing supervised classification learning algorithms*. Neural Computation, 10(7), 1895-1923.
- Government of South Africa. (2013). *Protection of Personal Information Act (POPIA), Act No. 4 of 2013*. Government Gazette of the Republic of South Africa.